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The Universal Standard Scale

Proposed Improvements to the American National Standards Institute (ANSI) Scale for Corneal Topography

Michael K. Smolek, PhD,¹ Stephen D. Klyce, PhD,¹ Jeffery K. Hovis, OD, PhD²

Purpose: To evaluate the American National Standards Institute (ANSI) corneal topography scale (ANSI Z-80.23-1999) and to develop a Universal Standard Scale (USS) based on scientific principles and clinical usefulness.

Design: Retrospective data analysis and computer modeling of frequency of corneal power distributions.

Methods: Four variations of corneal topography color scales were constructed using the ANSI standard description. The scales used 21 contour intervals with step sizes of 0.5, 1.0, or 1.5 diopters (D). In example 1, hue was varied, whereas saturation and brightness were held constant. In example 2, hue was varied systematically, whereas luminance was varied to compensate for spectral sensitivity. In example 3, hue was varied, whereas luminance was adjusted to improve color contrast at the center of the scale. In example 4, the palette was generated by varying the red, green, and blue color intensity levels. In the USS, colors were defined by hue, brightness, and saturation, and appearance was optimized by using contrast. The contour interval was determined from the standard deviation of keratometry for 27 normal corneas. The scale range was based on corneal powers found in 388 topography maps of 12 different corneal conditions.

Results: ANSI-derived maps were difficult to interpret because of poor color contrast and the inability to associate contours with specific colors in the scale. The 0.5 D interval slightly improved the appearance of some patterns but reduced the useful clinical range of the scale. ANSI colors did not match the user's expectations; abnormal patterns appeared in shades of green, which is associated with normality. The standard deviation of keratometry readings for the normal population was 1.59 D. Hence, the contour interval for the USS was set to 1.5 D. More than 99.9% of the corneal powers in 388 clinical maps were within the range of 30 to 67.5 D, which was chosen as the range for the USS.

Conclusions: The ANSI scale for corneal topography is a series of scales that lead to clinical misinterpretations. The alternative USS would provide uniform presentation, unambiguous interpretation, and greater ease of comparison among maps produced by different topographers. *Ophthalmology* 2002;109:361-369 © 2002 by the American Academy of Ophthalmology.

Color contour plots¹ are the preferred method of mapping corneal curvature, power, elevation, and, more recently, corneal and ocular wavefront optical path error. It has long been recognized that corneal topography displays should be constructed in a standardized fashion that is consistent with the user's expectations and needs.² Despite this, it has been difficult to form a consensus as to what that standard should be. At issue are the choice of contour interval size, range of powers to be displayed, and the design of the color palette;

together these parameters implement the cognitive function that permits clinical interpretation through pattern recognition and color association.

Improperly designed and loosely defined color scales can easily introduce confusion and errors of interpretation.³ For example, too small a step size for the contour interval will tend to emphasize clinically insignificant detail. Insufficient color contrast between adjacent contours masks clinically significant findings by making contours difficult to distinguish from one another. Poor color selection also makes it difficult to associate a contour's color to the respective color on a separate scale. In addition, multiple scales using identical colors to represent different corneal powers would make it difficult, if not impossible, to compare devices or communicate clinical findings with ease.

In 1999, the American National Standards Institute (ANSI) issued a report entitled *Corneal Topography Systems—Standard Terminology, Requirements (ANSI Z-80.23-1999)*.⁴ It was anticipated that this standard would allow users of corneal topographers to directly compare the topography maps produced by different manufacturers by use of a single color palette and specific fixed intervals. How-

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Table 1. American National Standards Institute (ANSI) Standard Corneal Topography Scales

Scale Interval (Diopters)	Mappable Corneal Powers (Diopters)		
	Lowest (Blue)	Midpoint (Green)	Highest (Red)
0.5	39	44	49
1.0	34	44	54
1.5	29	44	59

Adapted from Annex B. *Corneal Topography Systems—Standard Terminology, Requirements* (ANSI Z-80.23-1999).

ever, Annex B of the ANSI standard, which defines scale intervals, the scale center, and a color convention, does not specify a single, well-defined color palette but rather suggests a variety of numeric and color scale combinations.

Annex B states that “*standardized curvature maps shall use one of the three following corneal diopter intervals*”: 0.5, 1.0, and 1.5 diopters (D). The problem is that allowing the user a choice of one of three different scale intervals precludes the possibility of standardization, making it impossible to compare color topography maps from different machines. Furthermore, a scale interval could be selected on the basis of the visual impact of the display rather than the clinical basis for the interval.

The range selection seems to be even more open to interpretation than the interval size. The standard states that “*there should be at least 21 but not more than 25 displayed intervals differentiated by color*.” Table 1, adapted from a table in Annex B, indicates three different dioptric ranges that are potentially acceptable for 21-interval scales; however, the actual upper and lower limits are never precisely specified in the table or the text for intervals of more than 21 steps, so the total number of allowable standard scales is large.

With respect to assigning colors to the scale intervals, the document states that “*the hue shall be monotonically decreasing from green to red and shall be monotonically increasing from green to blue*.” Unfortunately, this statement is ambiguous in several ways. First, there are numerous ways to define “red,” “green,” and “blue” by color names alone, and these definitions can be variable both over time and among cultures.⁵ Second, the term “*monotonically*” could refer to a step interval in the color space defined by the coordinates of hue, saturation, and brightness (HSB color space), to equal steps of wavelength along the visible spectrum, or to an observer’s visual impression of equal steps of hue. Each interpretation could produce vastly different color scales. Furthermore, *monotonic* refers to a mathematical function that is progressively increasing or decreasing; it does not specify whether the function is linear or curvilinear.

A third ambiguity is that hue is ordinarily defined as that aspect of a color that changes most strongly with the wavelength of light, such as the bluishness or greenishness of a color.⁶ However, color is defined by more than hue alone. Color includes the attributes of saturation and brightness. Varying saturation and brightness, as well as hue, should be considered in designing the optimum color scale, because this permits enhancement of color contrast between adjacent

contours and improves the legibility of grayscale reproductions of the map.⁷ The ANSI standard does not mention varying brightness or saturation.

Finally, perceived color depends on the illumination and surrounding environment.⁷ Efforts to standardize color scales must take into account that viewing conditions can vary and make a color discrimination or color matching task more difficult.

In this study, we demonstrate the limitations of several variations of color scales that are based on different interpretations of the ANSI standard. We propose an alternative color scale, the Universal Standard Scale (USS), which was developed by determining the central power value, the optimum contour interval size, and the power range, and by carefully selecting colors for each contour interval to achieve optimum clinical usefulness. Examples of the ANSI standard scale are compared with the USS and to the previously validated Klyce-Wilson scale using typical clinical topography examples.

Material and Methods

Videokeratography

Corneal topography data files obtained with the TMS-1 (software version 1.6) and TMS-2 (software version 1.2) corneal topographers (Tomey Corp, Waltham, MA) were selected at random from more than 8000 TMS examinations from the medical records of the LSU Eye Center (New Orleans, LA) and the Southern Vision Institute (Dr Marguerite B. McDonald, New Orleans, LA). Informed consent was obtained from all patients before acquiring corneal topography. Custom programs were written in Visual Basic (Microsoft Corp, Redmond, WA) to read and analyze the keratography data files.

Color Display

For consistency and accuracy, Adobe Photoshop 5.5 for Windows (Adobe Systems, Inc, San Jose, CA) was used to sample colors and produce color scales. To compensate for the ambient lighting conditions, the Adobe Gamma Utility (Adobe Systems, Inc, San Jose, CA) was used to calibrate a new Trinitron flat-screen Ultrascan P991 color monitor (Dell Computer Corp, Round Rock, TX). The color temperature of the monitor was set to 9300° K, and the working profile was set to correct to 6500° K. The RGB color setup was set to sRGB color space, 6500° K white point, 2.2 Gamma (Microsoft Windows default). Color prints were made with an HP1220cxi color printer with PhotoRet III image technology enabled (Hewlett-Packard Corp, Palo Alto, CA).

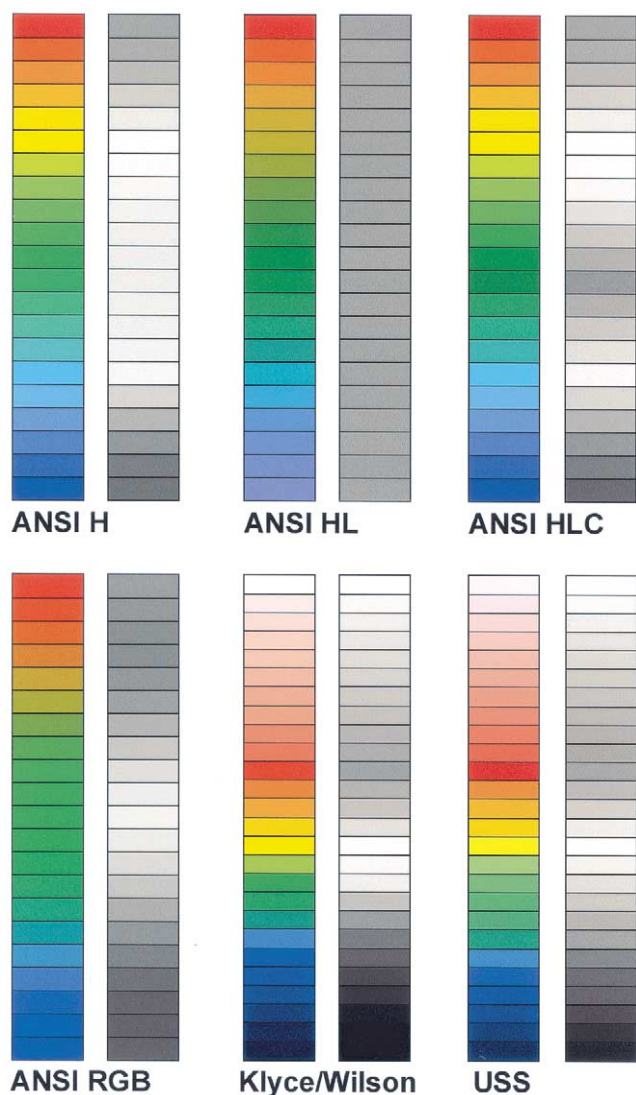


Figure 1. Color scales compared in this study: four different American National Standards Institute (ANSI) variations created by use of the description provided by the ANSI standard; the Klyce-Wilson scale; and the proposed Universal Standard Scale. Gray scale versions of the color scales illustrate how luminance contributes to improved contour discrimination.

ANSI Standard Scale Implementation

Several different examples of color scale variations that comply with the specifications given in the ANSI Standard Annex B were generated (Fig 1). Example 1 (for convenience abbreviated **ANSI H**) was defined using the HSB color space in which hue varies from 0 to 360° around the circumference of the HSB color space, whereas saturation and brightness potentially vary from 0% to 100%. Because the ANSI standard describes only monotonic steps of hue, both saturation and brightness were maintained at a constant 100% and hue was divided into 21 12-degree steps from 0 (red) to 240° (blue). It is important to note that although the brightness and saturation were set at 100%, these two attributes actually covary with hue and are not perceptually constant across hues.

Example 2 (**ANSI HL**) was similar to example 1, except that the luminance was held constant. This was accomplished by set-

ting the HSB coordinates to those of example 1 and then adjusting the L (the relative luminance) value of the CIE $L^*a^*b^*$ color space to produce a constant luminosity of 56% for all color steps. Fifty-six percent was chosen because it was the value of the initial pure red color interval when the HSB value was set to {0, 100, 100} and because it was a reasonable midrange value of luminance. Note that luminance is not constant for a constant brightness setting in the HSB system; it depends on the monitor phosphor output and the spectral sensitivity of the visual system.

Example 3 (**ANSI HLC**) was also based on example 1, except the brightness coordinate in HSB color space was graded in a V-shaped fashion in the central portion of the scale to increase contrast so as to better distinguish shades of green. This variation was not described by the ANSI standard document but seemed to be a reasonable method to enhance the color scale while remaining within the bounds of the ANSI definition.

Example 4 (**ANSI RGB**) of the ANSI color scale used RGB color space mixtures of the three primary colors: red {255, 0, 0}, green {0, 255, 0}, and blue {0, 0, 255}. Here, hue referred to the visual impression of the color and not to a strict definition of hue as in HSB color space. The Adobe Photoshop color gradient creation tool was used to automatically generate a linear spectrum of colors in which blue and red were manually set to the extreme endpoints of the gradient and green to the central position. The resulting image was then sampled with the eyedropper tool at 21 equidistant points along the gradient to obtain the RGB values used to specify the color palette of the scale.

Proposed USS

Numerical Scale Selection for the USS. The ideal contour interval would minimize irrelevant detail below a statistically defined threshold, such as 1 standard deviation (SD) from the mean corneal power of the global surface of normal corneas. The center of a topographic scale ideally should coincide with the mean of the normal distribution of powers obtained from the entire measurable surface area of normal corneas. Because topography is an important tool for discriminating between normal and abnormal corneas, different and distinct colors should be distributed across the entire range of powers encountered in the clinic.

To determine the central power value, 27 topographic maps of normal corneas were used. The powers from these maps were extracted, and a global mean and SD were determined.

The range of the scale was selected by making a frequency distribution of powers obtained from 388 topographic maps chosen to represent a variety of conditions: normal, clinical keratoconus, keratoconus suspect, contact lens-induced warpage, high astigmatism (≥ 1.5 D), laser in situ keratomileusis, penetrating keratoplasty, pellucid marginal degeneration, photorefractive keratectomy, and radial keratotomy. The numbers of maps included for each condition are shown in Table 2.

Palette Selection for the USS. A palette of colors was displayed on the color monitor and printed on premium inkjet glossy paper (HP C3836A; Hewlett-Packard Corp, Palo Alto, CA). These colors were adjusted subjectively by an individual (MKS) with normal color vision and extensive experience in the interpretation of topographic maps so as to enhance contrast and color distinction on both media outputs. Brightness, saturation, and hue were subject to manipulation. As hues reached their red and blue limits, steeper surfaces were characterized by progressively lighter shades of pinks, and flatter surfaces were characterized by progressively deeper or darker shades of blue. Violet and purple hues were not considered because they can be confused with their blue or red primaries. Similarly, brown hues were not used because they can be confused with orange or green, depending on the mixture of the red and green primaries.

Table 2. Global Average of Corneal Power by Condition

Condition	Mean	Standard Deviation	Number of Maps
Normal	42.76	1.59	27
High astigmatism	43.12	2.26	29
Contact lens-induced warpage	44.28	2.03	67
Keratoconus suspect	44.55	2.88	19
Keratoconus (mild)	45.03	3.65	22
Keratoconus (moderate)	47.77	6.14	31
Keratoconus (severe)	49.46	7.37	26
Keratoconus (combined)	47.49	6.23	79
Pellucid marginal degeneration	44.07	4.59	26
Penetrating keratoplasty	46.40	4.90	25
Laser in situ keratomileusis for myopia	41.22	2.38	42
Photorefractive keratectomy for myopia	41.98	2.88	41
Radial keratotomy	40.76	2.76	34
All conditions in data set	43.95	4.52	388

Results

The distribution of the normal cornea data for 27 maps is shown in Figure 2. The mean power was 42.76 with a SD of ± 1.59 D. This mean value was adjusted to 42.75 D to define the central value. The SD value of 1.59 D compared favorably with a topographic interval size of 1.50 D, a value in widespread use with corneal topographers, and the 1.50 D value was used to define the contour intervals. To associate color with normal and abnormal conditions, it was decided that corneal powers within ± 2 SD of the mean should have green colors. Subsequently, four contour intervals of green colors were placed with two intervals on either side of the mean. Powers in the interval immediately above 2 SD from the mean were indicated by yellow, and powers in the interval immediately below 2 SD from the mean were denoted by a cyan color.

The range of values that should be mapped distinctly by a topographic scale was determined to be 30.0 to 67.5 D (Fig 3). With this range, 99.9% of all values in the trial data set containing normal and abnormal corneas could be encompassed. Table 3

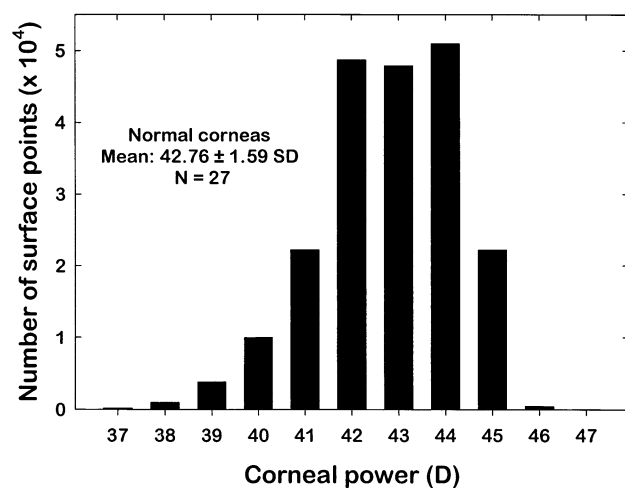


Figure 2. Histogram of normal corneal powers by corneal topography. Corneal power data were extracted from TMS files of 27 normal corneas and a frequency distribution made using 1-diopter (D) bins. The mean was 42.76 ± 1.59 D, with skewing toward lower powers because of peripheral flattening of the cornea.

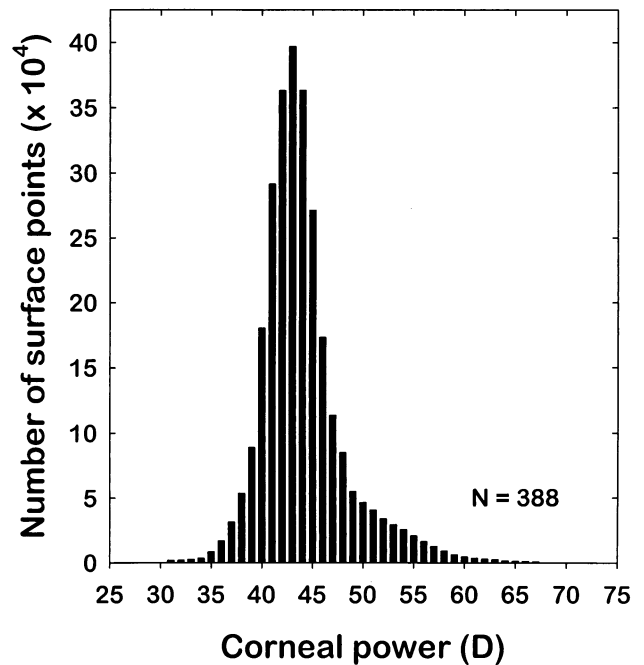


Figure 3. Histogram of clinical corneal surface powers by corneal topography. Corneal power data were extracted from TMS files of 388 corneas with 12 different conditions, and a frequency distribution was constructed using 1-diopter (D) bins. The most frequently occurring value was 43 D; 99.9% of all surface powers were contained in the range from 30 to 67.5 D. Note the skewing of the distribution toward higher powers, which is caused primarily by corneas with keratoconus.

indicates the clinical limitations of the ANSI standard. The 0.5-D scale was unable to map 13.8% of the upper and lower powers plotted in the frequency distribution in Figure 3. The 1.0-D scale also was hindered by an inadequate range.

Figure 1 shows the four variations of the ANSI standard scale, the Klyce-Wilson scale, and the proposed USS (reproduction of the colors in the figure may be slightly altered by the publication process). The ANSI variations tended to produce color scales that lacked distinct color differences between adjacent steps. In contrast, the Klyce-Wilson scale and the USS had easier-to-distinguish color contours, particularly in the important central portion of the scale, where distinctions between normal and abnormal powers are made. The exact color coordinates of the USS are provided in Table 4.

Table 3. Ability of Topography Scales to Display Corneal Powers

Scale Name	Interval Size (Diopters)	Range of Powers (Diopters)	Displayed Powers* (%)
ANSI	1.5	29–59	99.2
ANSI	1.0	34–54	96.6
ANSI	0.5	39–49	86.2
USS	1.5	30–67.5	99.9

*From 388 topographic maps of 12 corneal conditions (see Figure 2 and text for details).

ANSI = American National Standards Institute; USS = Universal Standard Scale.

Table 4. Universal Standard Scale

Color Name	sRGB	Hue, Saturation, Brightness (HSB)	Commission Internationale d'Eclairage L*a*b*	Diopters
Pinkish white	255, 238, 248	325, 7, 100	96, 7, -3	67.5
Light pink	255, 217, 227	344, 15, 100	90, 15, 0	66
Light pink	255, 197, 207	350, 23, 100	85, 22, 3	64.5
Light pink	255, 176, 187	352, 31, 100	80, 31, 7	63
Pink	255, 158, 168	354, 38, 100	76, 38, 11	61.5
Pink	255, 138, 148	355, 46, 100	71, 46, 16	60
Med pink	255, 115, 125	356, 55, 100	66, 55, 22	58.5
Med pink	255, 95, 105	356, 63, 100	63, 62, 29	57
Dark pink	255, 71, 80	357, 72, 100	59, 70, 39	55.5
Dark pink	255, 40, 50	357, 84, 100	56, 77, 52	54
Red	255, 0, 0	0, 100, 100	54, 81, 70	52.5
Dark orange	255, 102, 0	24, 100, 100	63, 57, 73	51
Medium orange	252, 153, 0	36, 100, 100	73, 33, 78	49.5
Yellow gold	252, 188, 0	45, 100, 99	80, 15, 82	48
Yellow	255, 255, 0	60, 80, 100	98, -15, 87	46.5
Light green	162, 250, 59	88, 76, 98	90, -48, 75	45
Medium green	80, 230, 51	110, 78, 90	81, -64, 68	43.5
Dark green	51, 204, 51	120, 75, 80	73, -61, 59	42
Cyan green	32, 176, 72	137, 82, 69	63, -54, 41	40.5
Cyan blue	0, 153, 102	160, 100, 60	56, -45, 17	39
Blue	0, 106, 157	199, 100, 62	42, -11, -36	37.5
Medium blue	0, 51, 204	255, 100, 80	29, 38, -83	36
Dark blue	0, 0, 204	240, 100, 80	22, 58, -95	34.5
Dark blue	0, 0, 153	240, 100, 60	15, 47, -77	33
Dark blue	0, 0, 112	240, 100, 44	9, 37, -61	31.5
Dark blue	0, 0, 80	240, 100, 31	4, 26, -47	30

The Universal Standard Scale uses the following video monitor profile settings: sRGB; Gamma 2.2; 6500 K; HDTV.

Figure 4 shows a cornea that has undergone myopic photorefractive keratectomy and has a central island mapped using six color scales: four ANSI variations with 1.5-D contour intervals, the Klyce-Wilson scale, and the USS. Note that the central island is less conspicuous and the irregular astigmatism less visible in all of the ANSI variations, because color selection was not optimized for contrast enhancement or designed to distinguish normal from abnormal powers. Manipulating luminance in the ANSI HLC example improved the legibility of individual color contours slightly, but the appearance of the Klyce-Wilson scale and the USS indicated that HSB should be optimized individually for each color interval to avoid confusion when matching the color contours in the map with the colors of the scale. In the USS, yellow denotes powers more than 2 SD from normal, and this color selection effectively highlights the central island.

Figure 5 shows the ANSI H scale using 0.5-, 1.0-, and 1.5-D intervals, as well as the proposed USS, for moderately high, with-the-rule corneal astigmatism. Note that the orientation and magnitude of the cylinder are difficult to appreciate using the 1.5-D ANSI H scale but much easier to interpret with the USS. The 0.5-D ANSI H scale does make the cylinder easy to see, but the contour pattern is noisy. The 1.0-D ANSI H scale superficially seems less noisy than the 0.5-D scale, but the green contours blend together. There is great difficulty in associating power with a specific color across all the ANSI standard scales; note that although the 44-D interval is consistently green across all ANSI variations, the red and blue extremes of the scales refer to completely different powers, depending on the interval size for the scale.

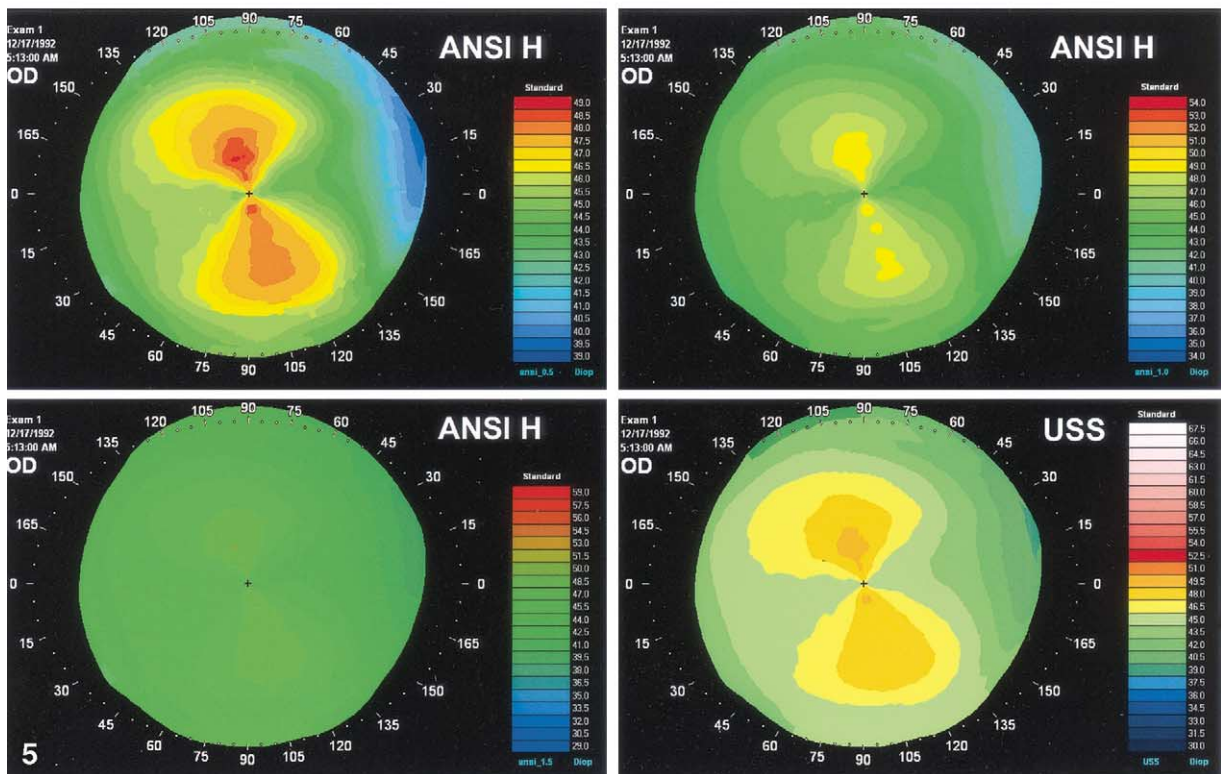
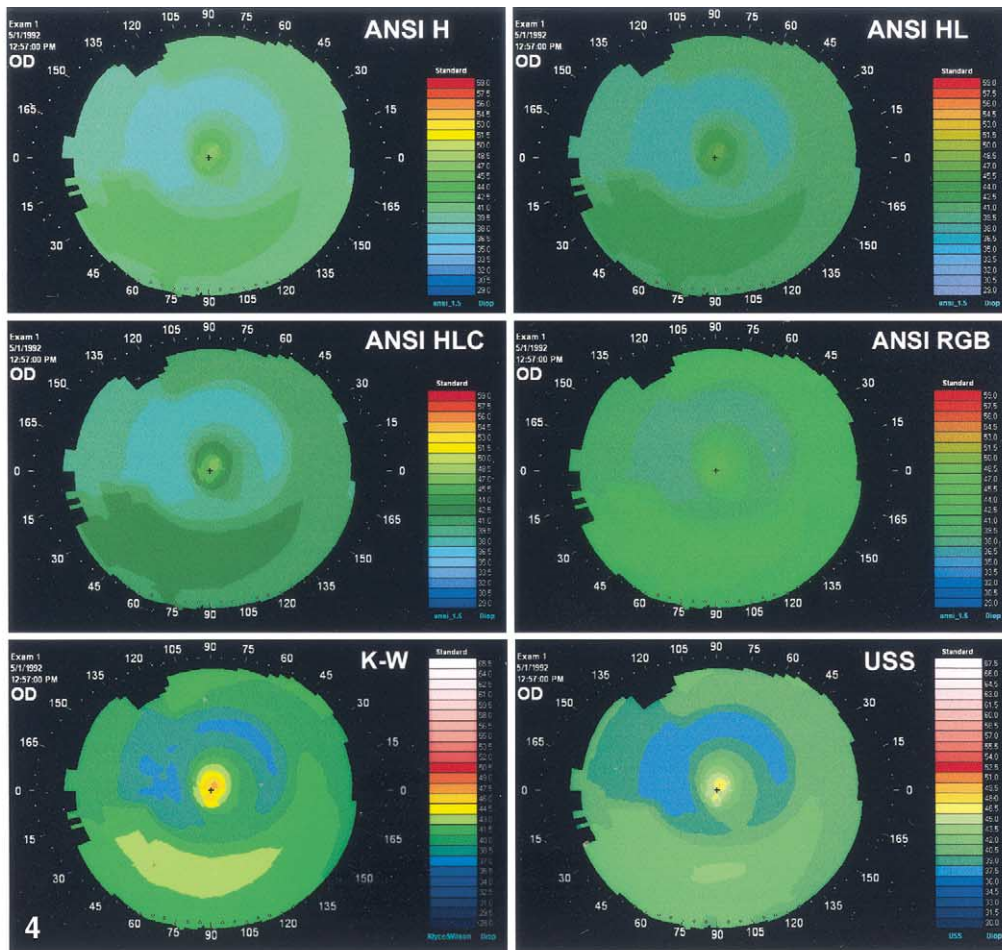
Figure 6 shows the ANSI H scale using 0.5-, 1.0-, and 1.5-D intervals, as well as the proposed USS, for a typical case of moderately severe keratoconus. Note that the severity level of the condition is difficult to appreciate using any of the ANSI standard

variations but easy to interpret with the USS. The 1.5-D ANSI H scale makes the keratoconus seem less severe than it actually is. The 0.5-D and 1.0-D ANSI H scales cannot map the steepest powers, because values greater than 49 or 54 D, respectively, are not identified by unique colors. Again, the 0.5- and 1.0-D scales are overly noisy, and the red and blue colors refer to different powers in the two scales. The USS performs as one expects a standard to perform: the entire range of unique powers in the map is displayed and the interpretation of the keratoconus condition and its severity is straightforward.

Discussion

On the basis of our results, the ANSI standard for the color scale is problematic for clinical reporting of corneal topography data. The ANSI standard permits a variety of different interval sizes, ranges, and color palettes, and, consequently, it does not provide for a single, true, standard display. Clinicians could be easily confused using a system of multiple interval sizes, ranges, and colors, whereas a single standard would provide consistent reporting of clinical data, including data in publications and presentations. A single display standard also would allow the output from different topography systems to be compared, and evidence of deficiencies in the underlying algorithms could be more easily documented. The ability to make comparisons among systems using standard test targets and a standard display was one of the primary interests of the ANSI standards committee.

To our knowledge, there is no scientific evidence to



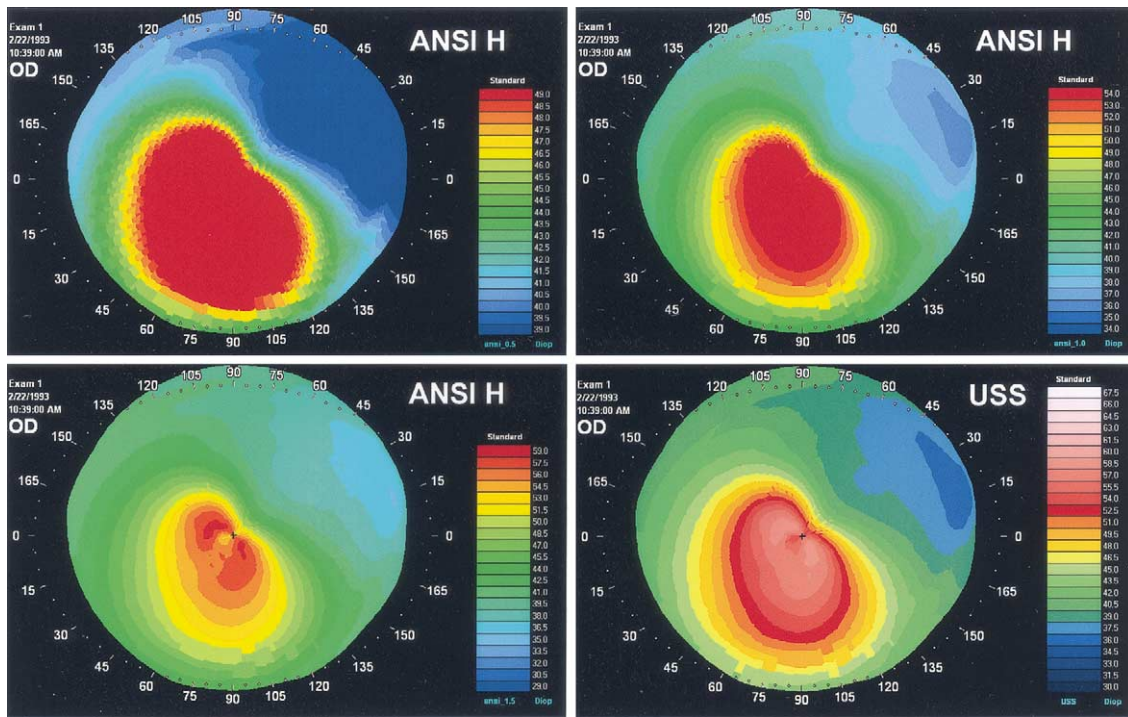


Figure 6. American National Standards Institute (ANSI) scale variations and the proposed Universal Standard Scale (USS) used to map a case of moderate keratoconus. All ANSI maps are in the ANSI H color scheme. **A**, ANSI scale with 0.5-diopter (D) intervals. The pattern is noisy, and the highest and lowest powers are off-scale. **B**, ANSI scale with 1.0-D intervals. The pattern is also noisy, and the highest powers remain off-scale. **C**, ANSI scale with 1.5-D intervals. The entire range of corneal powers is plotted, but the colors of the pattern are inconsistent with the severity level of keratoconus. **D**, The USS. Keratoconus is easy to interpret correctly.

support the idea that contour intervals smaller than 1.5 D improve clinical interpretation of corneal topography. On the contrary, Wilson et al² reported that, in a clinical study that used a commonly used corneal topographer, no topographic characteristics were present with a 1.0-D interval that were missed when examining maps with the 1.5-D interval. However, through our personal communication with numerous clinicians, we have been made aware that displaying maps with a 1.0- or 0.5-D interval often leads to misinterpretations of irregular astigmatism or early keratoconus in an otherwise normal cornea.

The 1.5-D contour interval was validated with the TMS-1 topographer (Tomey) and works well for other corneal topographers that have a similar spatial resolution ($\sim 140 \mu\text{m}$ between meridional samples on a 43-D surface). This resolution is determined by the distance between adjacent mires or their edges.

The ability to distinguish the range of normal corneal powers is the most demanding of topographic tasks, and the normal group has the smallest variance (Table 2). Consequently, the normal group should be used to define the contour interval. In this study, we found that the distribution of corneal powers in the group of normal corneas had a standard deviation of 1.59 D, which, for ease of use, was rounded to a value of 1.5 D. This value places an absolute lower limit on the contour step size needed to interpret maps correctly. This is not to say that other corneal power measures such as keratometry should be held to the same standard as measures used for interpreting global topography, because these measurements are entirely different from one another in terms of surface location, the extent of the surface being measured, and the application of the result.

We found that the range proposed by the ANSI standard (Table 1) did not allow complete coverage of the corneal

Figure 4. (Opposite) Four different interpretations of the American National Standards Institute (ANSI) standard, the Klyce/Wilson (K-W) scale, and the proposed Universal Standard Scale (USS) used to map a cornea with a central island after photorefractive keratectomy. All maps use 1.5-diopter intervals. **A**, ANSI H, with colors specified by hue varied systematically and saturation and brightness remaining constant. **B**, ANSI HL, with colors specified by hue varied systematically and luminance varied inversely to human spectral sensitivity to produce constant luminance. **C**, ANSI HLC, with colors specified by hue varied systematically and luminance artificially decreased in the center of the scale to enhance contrast. **D**, ANSI RGB, with colors specified by the Adobe Photoshop linear gradient color tool. **E**, The K-W absolute color scale. **F**, The USS. The central island is easily seen in both the K-W and the USS displays but is more difficult to visualize and the severity more difficult to judge in the ANSI variations.

Figure 5. The American National Standards Institute (ANSI) variations and the proposed Universal Standard Scale (USS) used to map astigmatism. All ANSI color scales are the ANSI H version. **A**, ANSI scale with 0.5-diopter (D) intervals. Cylinder orientation is obvious, but the contour pattern is noisy. **B**, ANSI scale with 1.0-D intervals. The pattern becomes less noisy, but colors are inconsistent with severity. **C**, ANSI scale with 1.5-D intervals. The contour pattern is not visible, and the green colors suggest a normal surface. **D**, The USS. The astigmatism is easy to discern, and the colors are logically related to power.

topography that one expects to encounter in a clinical setting (Fig 3). The ANSI standard indicates that the range can be extended in either direction from 21 to as many as 25 intervals, but this would make hue resolution of the additional colors, particularly green, even more difficult. We also found that the ANSI color palette was so loosely defined that every manufacturer could develop a unique color scale, unlike any other manufacturer's color scale, and yet remain in compliance with the ANSI standard.

We (MKS and SDK) voiced our concern to the ANSI standards draft committee about the color scale problem. The response was that the issue had no solution, because computer hardware varies widely in quality of output; and therefore, manufacturers should not be expected to standardize to a single, precise color scale. In fact, printing and computer graphics industry standards do exist and address the very issue of consistent color reproduction across computer systems using different hardware and software. These standards, which are regulated by the International Color Consortium and can be implemented across many operating systems, can be readily adapted for use with corneal topography displays. Algorithms with standard color specifications adopted by industry are readily available.⁸

The issue of calibration and exact color reproduction is critical for the ANSI standard, because all the ANSI color palettes we tested had very poor color resolution. In other words, a wide range of normal and abnormal powers were represented by nearly identical green-hued intervals that blended together on a typical ANSI color scale. By contrast, the USS has a built-in tolerance for monitors or printers that may not reproduce colors exactly (including liquid crystal diode video monitors) because the palette was selected on the basis of visual appearance and refined after testing on both calibrated and uncalibrated hardware. Implementing the USS color palette on a video system that is uncalibrated will produce inaccurate colors. However, the map and scale should still remain readable, because contrast also was used to help distinguish adjacent color steps.

There are more degrees of freedom using the perceptual attributes of HSB to specify color; therefore, it becomes easier to specify exact colors that can be displayed equally well across independent computer platforms compared with varying hue alone. A number of color systems have been designed to specify exact colors using precise coordinates in a color space. Color system examples include the Commission Internationale d'Eclairage (CIE) $L^*a^*b^*$, CIE XYZ, and standardized RGB (sRGB).⁸ CIE $L^*a^*b^*$ space was developed for industrial applications of color in an attempt to map perceptual differences into a more uniform space. The L refers to the axis for luminance, a refers to the red-green axis, and b refers to the blue-yellow axis. The CIE XYZ space is based on standardized color-matching data, where X, Y, Z represent the three idealized primary colors. RGB space refers to a color space formed by the additive mixture of the red, green, and blue phosphors in video displays. The output of each channel is scaled to intensity values between 0 and 255. sRGB is a device-independent version of RGB and is emerging as the preferred computer color graphics standard.

The dioptric values used to define the USS are provided in Table 4, along with the HSB coordinates, to demonstrate

how saturation and brightness can be covaried with hue to provide good color contrast between steps. The CIE $L^*a^*b^*$ values in Table 4 are exact for the monitor we used for development, but these values will differ with other video systems calibrated to different room lighting conditions. For consistent results, the device-independent sRGB coordinates in Table 4 should be used to duplicate the USS. The sRGB color coordinate system is the current Microsoft Windows and Hewlett-Packard color graphics standard.

With respect to color-deficient users, we were unable to do any rigorous testing, but we have anecdotal evidence from red-green deficient users that the contrast-adjusted contour steps in the USS are much easier to distinguish than either the ANSI scale variations or the Klyce/Wilson scale. Other than the USS, we know of no other topographic color scale that has been designed specifically to improve contour discrimination among all users, including color-deficient users.

In summary, the USS overcomes the problems encountered with the ANSI standard by associating a single, well-defined numerical scale with a single, well-defined color palette. This method produced maps that were consistent and could be rapidly and correctly interpreted. The USS displays a range of powers that encompasses 99.9% of both naturally occurring and surgically induced corneal shapes. The 1.5-D interval was shown to be the optimum interval size for general clinical use²; it also conforms closely to the standard deviation of normal corneal powers. This makes each map contour represent a 1-SD interval from the normal mean value near the center of the scale. The psychological impact of associating different colors with normal (green) and abnormally flat (cyan and blue) and steep (yellow, orange, and red) corneal powers¹ is preserved. By selectively choosing colors with specific HSB, the USS is understandable even when color reproduction is slightly off or when only a grayscale version is available. The specifications for the USS color scheme are presented using several different color space coordinates to aid manufacturers in reproducing the intended palette.

There are merits in adopting existing computer industry standards to optimize the use of color topography displays across different systems. The clinical use of topographic analysis would benefit if manufacturers would both use these standards and provide more assistance to end users in producing consistently high-quality output. This message should also be heeded by manufacturers of similar devices, such as systems that produce corneal elevation maps, ocular wavefront error maps, and pachymetry maps. Finally, we hope that the ANSI document can be amended to include the proposed USS or something similar that rigorously defines a single, effective color scale.

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